

Homework 4

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Math 23

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1 3.6

Question 6. Find the general solution of the given differential equation.

$$y'' + 4y' + 4y = t^{-2}e^{-2t}, \quad t > 0$$

Solution. The characteristic equation is $r^2 + 4r + 4 = 0 \implies (r + 2)^2 = 0 \implies r = -2$ (repeated root). Hence the homogeneous solution is

$$y_h = c_1e^{-2t} + c_2te^{-2t}.$$

So $y_1 = e^{-2t}$ and $y_2 = te^{-2t}$. We compute the Wronskian:

$$\begin{aligned} W(y_1, y_2) &= \begin{vmatrix} e^{-2t} & te^{-2t} \\ -2e^{-2t} & e^{-2t} - 2te^{-2t} \end{vmatrix} = e^{-2t}(e^{-2t} - 2te^{-2t}) - te^{-2t}(-2e^{-2t}) \\ &= e^{-4t} - 2te^{-4t} + 2te^{-4t} = e^{-4t}. \end{aligned}$$

By variation of parameters, with $g(t) = t^{-2}e^{-2t}$,

$$\begin{aligned} u_1 &= - \int \frac{y_2 g(t)}{W} dt = - \int \frac{te^{-2t} \cdot t^{-2}e^{-2t}}{e^{-4t}} dt = - \int \frac{t^{-1}e^{-4t}}{e^{-4t}} dt = - \int \frac{1}{t} dt = - \ln t, \\ u_2 &= \int \frac{y_1 g(t)}{W} dt = \int \frac{e^{-2t} \cdot t^{-2}e^{-2t}}{e^{-4t}} dt = \int \frac{t^{-2}e^{-4t}}{e^{-4t}} dt = \int t^{-2} dt = -t^{-1}. \end{aligned}$$

Therefore, the particular solution is

$$Y_p = u_1y_1 + u_2y_2 = (-\ln t)e^{-2t} + (-t^{-1})(te^{-2t}) = -e^{-2t} \ln t - e^{-2t}.$$

Since $-e^{-2t}$ is already part of the homogeneous solution, we drop it. Therefore the general solution is:

$$\boxed{y = c_1e^{-2t} + c_2te^{-2t} - e^{-2t} \ln t.}$$

□

Question 7. Find the general solution of the given differential equation.

$$4y'' + y = 2 \sec(t/2), \quad -\pi < t < \pi$$

Solution. The characteristic equation is $4r^2 + 1 = 0 \implies r = \pm \frac{i}{2}$. Hence the homogeneous solution is

$$y_h = c_1 \cos(t/2) + c_2 \sin(t/2).$$

So $y_1 = \cos(t/2)$ and $y_2 = \sin(t/2)$. We compute the Wronskian:

$$W(y_1, y_2) = \begin{vmatrix} \cos(t/2) & \sin(t/2) \\ -\frac{1}{2} \sin(t/2) & \frac{1}{2} \cos(t/2) \end{vmatrix} = \frac{1}{2} \cos^2(t/2) + \frac{1}{2} \sin^2(t/2) = \frac{1}{2}.$$

We must first write the equation in standard form by dividing by 4, showing

$$y'' + \frac{1}{4}y = \frac{1}{2} \sec(t/2),$$

so $g(t) = \frac{1}{2} \sec(t/2)$. By variation of parameters,

$$u_1 = - \int \frac{y_2 g(t)}{W} dt = - \int \frac{\sin(t/2) \cdot \frac{1}{2} \sec(t/2)}{\frac{1}{2}} dt = - \int \frac{\sin(t/2)}{\cos(t/2)} dt = - \int \tan(t/2) dt,$$

$$u_1 = 2 \ln |\cos(t/2)|.$$

$$u_2 = \int \frac{y_1 g(t)}{W} dt = \int \frac{\cos(t/2) \cdot \frac{1}{2} \sec(t/2)}{\frac{1}{2}} dt = \int \frac{\cos(t/2)}{\cos(t/2)} dt = \int 1 dt = t.$$

Therefore the particular solution is

$$Y_p = u_1 y_1 + u_2 y_2 = 2 \ln |\cos(t/2)| \cos(t/2) + t \sin(t/2),$$

and the general solution is

$$\boxed{y = c_1 \cos(t/2) + c_2 \sin(t/2) + 2 \cos(t/2) \ln |\cos(t/2)| + t \sin(t/2).}$$

□

Question 12. Verify that the given functions y_1 and y_2 satisfy the corresponding homogeneous equation; then find a particular solution of the given nonhomogeneous equation.

$$ty'' - (1+t)y' + y = t^2e^{2t}, \quad t > 0; \quad y_1(t) = 1+t, \quad y_2(t) = e^t$$

Solution. For $y_1 = 1+t$,

$$y_1' = 1, \quad y_1'' = 0.$$

Substituting into the homogeneous equation $ty'' - (1+t)y' + y = 0$:

$$t(0) - (1+t)(1) + (1+t) = 0 - (1+t) + (1+t) = 0.$$

For $y_2 = e^t$,

$$y_2' = e^t, \quad y_2'' = e^t.$$

Substituting gives

$$te^t - (1+t)e^t + e^t = e^t(t - 1 - t + 1) = 0.$$

Now we find the particular solution. We compute the Wronskian:

$$W(y_1, y_2) = \begin{vmatrix} 1+t & e^t \\ 1 & e^t \end{vmatrix} = (1+t)e^t - e^t = te^t.$$

We write the equation in standard form by dividing by t :

$$y'' - \frac{1+t}{t}y' + \frac{1}{t}y = te^{2t},$$

so $g(t) = te^{2t}$. By variation of parameters:

$$u_1 = - \int \frac{y_2 g(t)}{W} dt = - \int \frac{e^t \cdot te^{2t}}{te^t} dt = - \int e^{2t} dt = -\frac{e^{2t}}{2}.$$

$$u_2 = \int \frac{y_1 g(t)}{W} dt = \int \frac{(1+t) \cdot te^{2t}}{te^t} dt = \int (1+t)e^t dt.$$

Integrating by parts with $u = 1+t$ and $dv = e^t dt$:

$$u_2 = (1+t)e^t - \int e^t dt = (1+t)e^t - e^t = te^t.$$

Therefore the particular solution is

$$\begin{aligned} Y_p &= u_1 y_1 + u_2 y_2 = -\frac{e^{2t}}{2}(1+t) + te^t \cdot e^t \\ &= -\frac{(1+t)e^{2t}}{2} + te^{2t} = e^{2t} \left(-\frac{1}{2} - \frac{t}{2} + t \right) = \boxed{\frac{(t-1)e^{2t}}{2}}. \end{aligned}$$

□

2 7.1

Question 3. Transform the given equation into a system of first-order equations.

$$u^{(4)} - u = 0$$

Solution. We set $v_1 = u$, $v_2 = u'$, $v_3 = u''$, $v_4 = u'''$. Then,

$$\begin{aligned}v_1' &= v_2 \\v_2' &= v_3 \\v_3' &= v_4 \\v_4' &= u^{(4)} = u = v_1.\end{aligned}$$

In matrix form, this system can be written as:

$$\boxed{\begin{pmatrix} v_1' \\ v_2' \\ v_3' \\ v_4' \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} .}$$

□

Question 6. Systems of first-order equations can sometimes be transformed into a single equation of higher-order. Consider the system

$$x_1' = -2x_1 + x_2, \quad x_2' = x_1 - 2x_2.$$

- (a) Solve the first differential equation for x_2 .
- (b) Substitute the result of (a) into the second differential equation, thereby obtaining a second-order differential equation for x_1 .
- (c) Solve the differential equation found in (b) for x_1 .
- (d) Use the results of (a) and (c) to find x_2 .

Solution. For part (a), solving the first equation $x_1' = -2x_1 + x_2$ for x_2 gives

$$\boxed{x_2 = x_1' + 2x_1.}$$

For part (b),

$$x_2' = x_1'' + 2x_1',$$

so substituting x_2 and x_2' into the second equation $x_2' = x_1 - 2x_2$ results in

$$x_1'' + 2x_1' = x_1 - 2(x_1' + 2x_1)$$

$$x_1'' + 2x_1' = x_1 - 2x_1' - 4x_1$$

$$\boxed{x_1'' + 4x_1' + 3x_1 = 0.}$$

For part (c), the characteristic equation is

$$r^2 + 4r + 3 = 0 \implies (r + 1)(r + 3) = 0 \implies r = -1, -3.$$

Therefore,

$$\boxed{x_1 = c_1 e^{-t} + c_2 e^{-3t}.}$$

This makes part (d) trivial because

$$x_2 = x_1' + 2x_1$$

$$= (-c_1 e^{-t} - 3c_2 e^{-3t}) + 2(c_1 e^{-t} + c_2 e^{-3t}) = c_1 e^{-t} - 3c_2 e^{-3t} + 2c_1 e^{-t} + 2c_2 e^{-3t}$$

$$\boxed{x_2 = c_1 e^{-t} - c_2 e^{-3t}.}$$

□

Question 13. Let $x = x_1(t)$, $y = y_1(t)$ and $x = x_2(t)$, $y = y_2(t)$ be any two solutions of the linear nonhomogeneous system

$$x' = p_{11}(t)x + p_{12}(t)y + g_1(t),$$

$$y' = p_{21}(t)x + p_{22}(t)y + g_2(t).$$

Show that $x = x_1(t) - x_2(t)$, $y = y_1(t) - y_2(t)$ is a solution of the corresponding homogeneous system.

Proof. Since $x = x_1(t)$ and $y = y_1(t)$ is a solution of the system, we have

$$\begin{cases} x'_1 = p_{11}(t)x_1 + p_{12}(t)y_1 + g_1(t) \\ y'_1 = p_{21}(t)x_1 + p_{22}(t)y_1 + g_2(t). \end{cases}$$

Since $x = x_2(t)$ and $y = y_2(t)$ is also a solution, we have

$$\begin{cases} x'_2 = p_{11}(t)x_2 + p_{12}(t)y_2 + g_1(t) \\ y'_2 = p_{21}(t)x_2 + p_{22}(t)y_2 + g_2(t). \end{cases}$$

Now let $x = x_1(t) - x_2(t)$ and $y = y_1(t) - y_2(t)$. Subtracting the second pair of equations from the first gives

$$\begin{cases} x'_1 - x'_2 = p_{11}(t)(x_1 - x_2) + p_{12}(t)(y_1 - y_2) + \cancel{g_1(t)} - \cancel{g_1(t)} \\ y'_1 - y'_2 = p_{21}(t)(x_1 - x_2) + p_{22}(t)(y_1 - y_2) + \cancel{g_2(t)} - \cancel{g_2(t)}. \end{cases}$$

$$\begin{cases} (x_1 - x_2)' = p_{11}(t)(x_1 - x_2) + p_{12}(t)(y_1 - y_2) \\ (y_1 - y_2)' = p_{21}(t)(x_1 - x_2) + p_{22}(t)(y_1 - y_2). \end{cases}$$

Therefore $x = x_1(t) - x_2(t)$ and $y = y_1(t) - y_2(t)$ satisfy the corresponding homogeneous system

$$x' = p_{11}(t)x + p_{12}(t)y,$$

$$y' = p_{21}(t)x + p_{22}(t)y.$$

□

3 7.2

Question 16. If $\mathbf{A}(t) = \begin{pmatrix} e^t & 2e^{-t} & e^{2t} \\ 2e^t & e^{-t} & -e^{2t} \\ -e^t & 3e^{-t} & 2e^{2t} \end{pmatrix}$ and $\mathbf{B}(t) = \begin{pmatrix} 2e^t & e^{-t} & 3e^{2t} \\ -e^t & 2e^{-t} & e^{2t} \\ 3e^t & -e^{-t} & -e^{2t} \end{pmatrix}$, find

(a) $\mathbf{A} + 3\mathbf{B}$

(b) $\mathbf{AB}$

(c) $\frac{d\mathbf{A}}{dt}$

(d) $\int_0^1 \mathbf{A}(t) dt$

Solution. For part (a),

$$\begin{aligned} \mathbf{A} + 3\mathbf{B} &= \begin{pmatrix} e^t & 2e^{-t} & e^{2t} \\ 2e^t & e^{-t} & -e^{2t} \\ -e^t & 3e^{-t} & 2e^{2t} \end{pmatrix} + 3 \begin{pmatrix} 2e^t & e^{-t} & 3e^{2t} \\ -e^t & 2e^{-t} & e^{2t} \\ 3e^t & -e^{-t} & -e^{2t} \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 7e^t & 5e^{-t} & 10e^{2t} \\ -e^t & 7e^{-t} & 2e^{2t} \\ 8e^t & 0 & -e^{2t} \end{pmatrix}}. \end{aligned}$$

For part (b), computing each entry of $\mathbf{AB}$ using row times column:

$$\mathbf{AB} = \boxed{\begin{pmatrix} 2e^{2t} - 2 + 3e^{3t} & 1 + 4e^{-2t} - e^t & 3e^{3t} + 2e^t - e^{4t} \\ 4e^{2t} - 1 - 3e^{3t} & 2 + 2e^{-2t} + e^t & 6e^{3t} + e^t + e^{4t} \\ -2e^{2t} - 3 + 6e^{3t} & -1 + 6e^{-2t} - 2e^t & -3e^{3t} + 3e^t - 2e^{4t} \end{pmatrix}}.$$

For part (c), differentiating each entry of $\mathbf{A}(t)$:

$$\frac{d\mathbf{A}}{dt} = \boxed{\begin{pmatrix} e^t & -2e^{-t} & 2e^{2t} \\ 2e^t & -e^{-t} & -2e^{2t} \\ -e^t & -3e^{-t} & 4e^{2t} \end{pmatrix}}.$$

For part (d), integrating each entry of $\mathbf{A}(t)$ from 0 to 1:

$$\begin{aligned} \int_0^1 \mathbf{A}(t) dt &= \begin{pmatrix} \int_0^1 e^t dt & \int_0^1 2e^{-t} dt & \int_0^1 e^{2t} dt \\ \int_0^1 2e^t dt & \int_0^1 e^{-t} dt & \int_0^1 -e^{2t} dt \\ \int_0^1 -e^t dt & \int_0^1 3e^{-t} dt & \int_0^1 2e^{2t} dt \end{pmatrix} \\ &= \boxed{\begin{pmatrix} e - 1 & 2(1 - e^{-1}) & \frac{e^2 - 1}{2} \\ 2(e - 1) & 1 - e^{-1} & -\frac{e^2 - 1}{2} \\ -(e - 1) & 3(1 - e^{-1}) & e^2 - 1 \end{pmatrix}}. \end{aligned}$$

□

Question 20. Verify that the given matrix satisfies the given differential equation.

$$\Psi' = \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} \Psi, \quad \Psi(t) = \begin{pmatrix} e^t & e^{-2t} & e^{3t} \\ -4e^t & -e^{-2t} & 2e^{3t} \\ -e^t & -e^{-2t} & e^{3t} \end{pmatrix}$$

Solution. On the left hand side,

$$\Psi'(t) = \frac{d}{dt} \begin{pmatrix} e^t & e^{-2t} & e^{3t} \\ -4e^t & -e^{-2t} & 2e^{3t} \\ -e^t & -e^{-2t} & e^{3t} \end{pmatrix} = \begin{pmatrix} e^t & -2e^{-2t} & 3e^{3t} \\ -4e^t & 2e^{-2t} & 6e^{3t} \\ -e^t & 2e^{-2t} & 3e^{3t} \end{pmatrix}.$$

On the right hand side,

$$\begin{aligned} \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} \Psi &= \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} e^t & e^{-2t} & e^{3t} \\ -4e^t & -e^{-2t} & 2e^{3t} \\ -e^t & -e^{-2t} & e^{3t} \end{pmatrix} \\ &= \begin{pmatrix} e^t + 4e^t - 4e^t & e^{-2t} + e^{-2t} - 4e^{-2t} & e^{3t} - 2e^{3t} + 4e^{3t} \\ 3e^t - 8e^t + e^t & 3e^{-2t} - 2e^{-2t} + e^{-2t} & 3e^{3t} + 4e^{3t} - e^{3t} \\ 2e^t - 4e^t + e^t & 2e^{-2t} - e^{-2t} + e^{-2t} & 2e^{3t} + 2e^{3t} - e^{3t} \end{pmatrix} \\ &= \begin{pmatrix} e^t & -2e^{-2t} & 3e^{3t} \\ -4e^t & 2e^{-2t} & 6e^{3t} \\ -e^t & 2e^{-2t} & 3e^{3t} \end{pmatrix} \end{aligned}$$

Therefore, both sides are equal, and the matrix satisfies the differential equation. □

4 7.3

Question 4. Either solve the given system of equations, or else show that there is no solution.

$$x_1 + 2x_2 - x_3 = 0$$

$$2x_1 + x_2 + x_3 = 0$$

$$x_1 - x_2 + 2x_3 = 0$$

Solution. We rewrite the system of equations in the form

$$\begin{pmatrix} 1 & 2 & -3 \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \mathbf{0}.$$

Let us reduce the augmented coefficient matrix into RREF:

$$\begin{aligned} & \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & -1 & 2 & 0 \end{pmatrix} \xrightarrow{R_2-2R_1, R_3-R_1} \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & -3 & 3 & 0 \end{pmatrix} \\ & \xrightarrow{R_3-R_2} \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{-\frac{1}{3}R_2} \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ & \xrightarrow{R_1-2R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Therefore, the system has infinitely many solutions of the form

$$\boxed{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = t \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.}$$

□

Question 12. Determine whether the members of the given set of vectors are linearly independent for $-\infty < t < \infty$. If they are linearly dependent, find the linear relation among them.

$$\mathbf{x}^{(1)}(t) = (2 \sin t, \sin t), \quad \mathbf{x}^{(2)}(t) = (\sin t, 2 \sin t)$$

Solution. If the two vectors were linearly dependent, then one must be a scalar multiple of the other. Suppose they were scalar multiples of each other. Then, $\mathbf{x}^{(1)} = k\mathbf{x}^{(2)}$ for some constant k . However,

$$2 \sin t = k \sin t \implies k = 2 \quad \text{and} \quad \sin t = 2k \sin t \implies k = \frac{1}{2},$$

which is a contradiction. Therefore the vectors are linearly independent for $-\infty < t < \infty$. $\square$

Question 13. Let

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} e^t \\ te^t \end{pmatrix}, \quad \mathbf{x}^{(2)}(t) = \begin{pmatrix} 1 \\ t \end{pmatrix}.$$

Show that $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(2)}(t)$ are linearly dependent at each point in the interval $0 \leq t \leq 1$. Nevertheless, show that $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(2)}(t)$ are linearly independent on $0 \leq t \leq 1$.

Proof. For the first part, at any point, $t_0 \in [0, 1]$,

$$\mathbf{x}^{(1)}(t_0) = \begin{pmatrix} e^{t_0} \\ t_0 e^{t_0} \end{pmatrix} = e^{t_0} \begin{pmatrix} 1 \\ t_0 \end{pmatrix} = e^{t_0} \mathbf{x}^{(2)}(t_0).$$

Since $\mathbf{x}^{(1)}(t_0) = e^{t_0} \mathbf{x}^{(2)}(t_0)$ at every fixed point t_0 , the two vectors are linearly dependent at each point in $0 \leq t \leq 1$. Now we show the latter. Suppose there exist constants c_1 and c_2 such that

$$c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}(t) = \mathbf{0} \quad \text{for all } t \in [0, 1].$$

This gives the system

$$\begin{aligned} c_1 e^t + c_2 &= 0 \\ c_1 t e^t + c_2 t &= 0. \end{aligned}$$

From the first equation,

$$c_2 = -c_1 e^t.$$

But this equation contradicts because c_2 must be a constant. Therefore $c_1 = 0$, and substituting back into the first equation gives $c_2 = 0$ as well. Hence, the vectors are linearly independent on $0 \leq t \leq 1$. $\square$

Question 20. Find all eigenvalues and eigenvectors of the given matrix.

$$\begin{pmatrix} \frac{11}{9} & -\frac{2}{9} & \frac{8}{9} \\ \frac{2}{9} & \frac{2}{9} & \frac{10}{9} \\ \frac{8}{9} & \frac{10}{9} & \frac{5}{9} \end{pmatrix}$$

Solution. To simplify, we factor out $\frac{1}{9}$ and work with $9A$ as such:

$$9A = \begin{pmatrix} 11 & -2 & 8 \\ -2 & 2 & 10 \\ 8 & 10 & 5 \end{pmatrix}.$$

The characteristic polynomial of A is $\det(A - \lambda I) = 0$. Multiplying through by 9, this is equivalent to $\det(9A - 9\lambda I) = 0$:

$$\det \begin{pmatrix} 11 - 9\lambda & -2 & 8 \\ -2 & 2 - 9\lambda & 10 \\ 8 & 10 & 5 - 9\lambda \end{pmatrix} = 0.$$

The left hand sides evaluates as such:

$$(11 - 9\lambda)[(2 - 9\lambda)(5 - 9\lambda) - 100] + 2[-2(5 - 9\lambda) - 80] + 8[-20 - 8(2 - 9\lambda)] = 0.$$

$$(11 - 9\lambda)(81\lambda^2 - 63\lambda - 90) + 2(18\lambda - 90) + 8(72\lambda - 36) = 0$$

$$-729\lambda^3 + 1458\lambda^2 + 729\lambda - 1458 = 0$$

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0.$$

$$\lambda^2(\lambda - 2) - (\lambda - 2) = 0 \implies (\lambda^2 - 1)(\lambda - 2) = 0 \implies \boxed{\lambda_1 = 1, \quad \lambda_2 = -1, \quad \lambda_3 = 2.}$$

For $\lambda_1 = 1$, we solve $(A - I)\mathbf{v} = \mathbf{0}$, equivalently $(9A - 9I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 2 & -2 & 8 \\ -2 & -7 & 10 \\ 8 & 10 & -4 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = -1$, we solve $(A + I)\mathbf{v} = \mathbf{0}$, equivalently $(9A + 9I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 20 & -2 & 8 \\ -2 & 11 & 10 \\ 8 & 10 & 14 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(2)} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}.$$

For $\lambda_3 = 2$, we solve $(A - 2I)\mathbf{v} = \mathbf{0}$, equivalently $(9A - 18I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} -7 & -2 & 8 \\ -2 & -16 & 10 \\ 8 & 10 & -13 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(3)} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}.$$

Therefore the eigenvalues and corresponding eigenvectors are:

$$\lambda_1 = 1, \mathbf{v}^{(1)} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}; \quad \lambda_2 = -1, \mathbf{v}^{(2)} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}; \quad \lambda_3 = 2, \mathbf{v}^{(3)} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}.$$

□